Deadlines You Can Docket Against

Verifying Computed Trademark Deadlines Against 300,000 Register Records

Signa Research

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Abstract

Most trademark data errors cost analysis quality. Deadline errors cost the trademark. A missed renewal or maintenance filing does not degrade a registration; it cancels it. The calendars that prevent this (docketing systems) are traditionally maintained by hand, office by office, at considerable expense. Signa computes these deadlines: renewal cycles, declarations of use, grace periods, restoration windows, and opposition periods, from statutory rules modeled for 22 jurisdictions and exposed directly in the API.

This report does two things. First, it describes the engine and the standard we hold it to: deadline computation at Signa is a pure function (no database, no network, no clock dependence beyond an explicit as-of date), every rule carries citations to its statutory sources with a last-verified date, and the full rule corpus is itself published through the API for inspection. Second, and mainly, it reports a register-based verification. We froze a pre-registered, hash-manifested sample of 31,547 registrations across 10 production offices, then an expanded same-seed sample (stability check) of 306,896 that is a strict superset of the first (the same seed, so every original record is retained), and measured our computed deadlines against three complementary forms of ground truth: dates the offices themselves publish on the record (for one office, USPTO, the compared date is connector-derived; that cell is flagged and its independent evidence is the event arm), the dated renewal and maintenance filings that actually occurred in register histories, and 34,132 observed opposition proceedings. The rules were corrected against these same records during the evaluation, so the expanded sample is a stability check, not an independent holdout.

On the full population, exact agreement between the computed renewal deadline and the office-published date is 96.08% of 282,393 comparisons at expanded scale (96.00% on the committed small sample). Weighted by each stratum's true production share, exact agreement for the pure statutory computation is 98.55%. On a modeled population defined after mechanism analysis (post hoc), which excludes three documented subpopulations (Australian direct marks filed before 1996, Singapore direct marks filed before 1999, and Madrid subsequent designations whose anchoring date is absent from the row), exact agreement is 99.74% of 282,393 comparisons (95% CI 99.72 to 99.76). A production-weighted figure of 99.89% describes the schedule the API actually serves; it is office-date-anchored and therefore partially self-referential, and is not evidence of rule correctness. 93.1% of 27,562 observed renewal and maintenance filings fall inside our computed windows, and 90.2% of 23,751 verifiable opposition filings fall inside our computed opposition windows (98.1% at EUIPO, the cleanest cell). Every disagreement was individually classified by mechanism; after that classification, the genuinely unexplained residual is 3 records out of 282,393 comparisons, or 0.001%.

The verification also found defects: seven in our own rules (six in the maintenance rules, one in the opposition rules), all fixed, with agreement reported before and after each fix; three in our data ingestion (two fixed, one filed); and 311 records classified by Signa as register-data errors, with per-record evidence, where the register rather than the computation carries the wrong date. We publish the methodology in full, including the discrepancy taxonomy and the cases we got wrong, because a deadline product asks for a specific kind of trust, and we believe that trust should be checkable.

About this report. This is a first-party evaluation: Signa selected the metrics, wrote the evaluator, corrected the system under test, and classified the disagreements. It has not been independently audited. The artifacts are published so that any reader can check the work.

1. Background: the docketing problem

Registered trademarks are not granted once; they are kept alive. Nearly every jurisdiction requires periodic renewal, and several require affirmative proof that the mark is actually in use: the United States demands a declaration of use between the fifth and sixth year after registration (Section 8 of the Lanham Act), again at each renewal (Section 8 combined with Section 9), and under a separate provision for Madrid-route registrations (Section 71). Miss the window, and after a short grace period the registration is cancelled. Oppositions run on even tighter calendars: a third party typically has a fixed window after publication to oppose, and a brand owner watching for conflicts has exactly that window to act.

What makes this hard to automate is not any single rule but the accumulation of jurisdictional texture:

- **Different triggers.** Renewal terms run from filing in some jurisdictions and from registration in others. Sweden switched triggers on 2019-01-01, and the correct trigger for a Swedish mark depends on which side of that reform it was granted.
- **Route matters.** The same US registration is governed by Section 8 or Section 71 depending on whether it arrived directly or as a Madrid designation, and a Madrid designation's renewal anywhere runs on the international registration's ten-year cycle, not the national trigger.
- **Legacy regimes.** Canadian registrations renewed before the 2019-06-17 reform carry 15-year terms and can sit on mixed 15-and-10-year ladders. Australian marks accepted before 2013-04-15 had a three-month opposition window; later ones have two.
- **Calendar arithmetic is legally defined.** What "three months after" means at a month boundary is specified by statute and differs by country (German law, the Canadian Interpretation Act, the Singapore Interpretation Act, and EU delegated regulation all define it, not identically). Leap days, month-end clamping, end-of-month rules for older German registrations, and weekend or holiday rollover (37 CFR 2.196 in the US, Singapore's excluded-day practice including the eves of three holidays) all change real dates.
- **A deadline is a legal day, not a timestamp.** An opposition window closes at the end of the office's legal day in the office's time zone. A system that treats it as a UTC timestamp will tell a California user that a European deadline closed hours before it legally did.

Law firms and corporate IP departments solve this with commercial docketing systems and manual verification. Developers building trademark features have had no API-native option: general trademark data APIs return raw dates from the register and leave the statutory arithmetic to the caller.

1.1 Why verification, not just testing

The engine described here is extensively unit-tested against hand-derived golden dates (section 3.4). But hand-derived goldens share a failure mode with the rules they test: both are produced by humans reading statutes. The strongest available check is the register itself: offices publish expiry and renewal-due dates on records, register histories contain the dated renewal and maintenance filings that actually happened, and opposition proceedings carry the dates oppositions were actually filed. If our computed deadline for a real registration disagrees with what the office printed on that registration, one of us is wrong, and either answer to that question is worth publishing. That comparison, at scale, across offices, is this report.

2. What is computed

The engine produces two families of output.

Maintenance deadlines. For a given mark (jurisdiction, filing route, status, and the record's date fields), the engine returns the schedule of lifecycle deadlines: renewal (recurring by statutory cycle, including legacy-term ladders where a reform changed the cycle length), declaration of use (US Sections 8 and 71), combined renewal and use filings (US Section 8 with Section 9), declaration of incontestability (US Section 15, the one optional entry), restoration windows following a missed deadline, and international-registration renewal for Madrid records. Each computed deadline carries the date the window opens, the due date, the grace-period expiry where one exists, a status relative to an explicit as-of date (future, window open, due soon, in grace, missed), an urgency tier, and the consequence if missed (cancellation, expiration, or lapse of the international registration).

Opposition windows. For a published application or registration, the engine computes the statutory opposition window: trigger event (publication for opposition, registration publication, or international-designation publication, depending on the office), open and close dates, the routinely available extension where office practice has one, and a must-act-by moment expressed as the end of the office's legal day in the office's time zone with daylight saving handled. Each computed window carries the citation and version of the rule that produced it.

3. The engine

3.1 Rules as audited data, not buried code

Rules are data: 23 jurisdiction configurations covering 22 jurisdictions for maintenance deadlines (the United States carries two, one for direct registrations and one for Madrid designations), and 27 opposition-window rules covering both direct and Madrid-designation routes at the covered offices. Every configuration carries citations to its statutory and office sources, each with the date it was last checked, and a last-verified date for the configuration as a whole. Computed opposition windows embed the rule version that produced them, so any alert or API response can be traced to the exact rule text it came from.

The rule corpus is not a secret. It is published, with citations, through the API (`GET /v1/deadline-rules` and `GET /v1/opposition-rules`), so a customer or their counsel can audit the rules we apply before relying on them. The full deadline rule corpus is published for inspection.

3.2 Purity as a correctness strategy

Deadline computation is a pure function: no database access, no network calls, and no hidden dependence on the current clock (callers pass an explicit as-of date; statuses and urgencies are computed relative to it). Identical inputs always produce identical output. This is a correctness strategy, not an implementation detail: pure functions can be property-tested exhaustively, reproduced exactly in an investigation, and recomputed identically at read time, so there is no stored deadline to go stale when a rule is corrected.

Dates are treated as calendar dates, not timestamps, except at the one place time zones legally matter: the must-act-by moment of an opposition window, which is computed as the end of the office's legal day in the office's own time zone. Date arithmetic follows the applicable legal definitions: month addition clamps to month-end where statutes say so, leap days resolve the way the relevant interpretation acts specify, older German registrations get the end-of-month rule, pre-reform Canadian registrations get the legacy-term ladder, the Swedish trigger change applies from its statutory date keyed on the registration date, and deadlines roll forward over weekends and office holidays against 24 office holiday calendars maintained for 2024 through 2030 (including, for Singapore, the office's excluded-day eves).

3.3 The office's own dates as an anchor

Two behaviors connect the statutory computation to what the office has already said about a record. First, when the office states an expiry date and the statutory ladder computed from the record's trigger date passes through it exactly, the office has confirmed the computation and the full trigger-derived schedule is used. When the stated expiry does not sit on the computable ladder (because a discriminating date is absent from the record, as with Madrid subsequent designations, or because a legacy cadence predates the modeled regime), the schedule is re-anchored on the office's stated date. Second, a renewal cycle that fell due before the office-stated expiry has by definition already been paid (that is why the term now runs to the stated expiry), so it is not shown as an actionable deadline. Both behaviors are measured separately in section 5: the pure statutory computation is evaluated with the office's dates withheld, precisely so that the evaluation cannot be circular.

3.4 Exposure and cost

Computed deadlines appear on the trademark detail endpoint (`GET /v1/trademarks/{id}`), on portfolio deadline views (including an iCalendar feed), and inside watch alerts, where the opposition must-act-by is computed with the provenance fields attached. For bulk and integration use, two dedicated endpoints compute without persisting anything: `POST /v1/deadlines/compute` and `POST /v1/oppositions/compute` accept up to 1,000 items per call, take an explicit as-of date and horizon, and return per-item results including an explicit `supported: false` with a reason for jurisdictions we have not modeled, rather than a silent guess. The engine computes full schedules for the entire 31,547-record evaluation sample in roughly 256 milliseconds single-threaded on a laptop, about 8 microseconds per record.

3.5 Standing tests and the frozen regression gate

The rules ship with approximately 346 test cases in the rules package: per-office golden dates for maintenance and opposition computations, calendar-arithmetic and holiday-rollover suites, and property-based tests executed across every jurisdiction configuration. In addition, a frozen regression gate runs in continuous integration on every change: it recomputes the 31,547-record maintenance sample and pins that sample's Arm A and Arm B agreement totals only, so an engine change that moves a single record among those 31,547 fails the build until it is re-verified cell by cell. The gate does not cover the expanded 306,896-record run, the production-weighted outputs, or the opposition arm.

4. Verification methodology

4.1 Design principles

Four commitments govern the design. The sample was frozen (record identifier lists hashed with SHA-256) before any comparison ran. Every number carries its n, headline numbers carry 95% confidence intervals (Wilson), and no cell below n=50 is reported. Every disagreement is individually classified by mechanism, not left in an "other" bucket. And when the evaluation itself exposed a defect in a rule, we corrected the rule and report agreement both before and after the correction. No correction was accepted on empirical fit alone; each required a statutory or documented-data-source justification, verified independently against primary sources, and was required to fit the register data without regressing any other cell. Several implementation choices (anchor selection, era boundaries) were informed by the data and are disclosed as such. Correcting rules against the evaluation and then reporting only the corrected number would make the exercise circular; showing both makes a correction evidence of the process working. One protocol amendment is disclosed rather than left silent: the pre-registered comparison horizon of 15 years was raised to 45 years during the evaluation after it produced false disagreements on old Canadian records (section 5.7); a wider horizon cannot create false matches.

4.2 Samples

Three frozen datasets are used.

- **Maintenance sample.** A deterministic, seeded sample of 31,547 registrations across 10 production offices (USPTO, EUIPO, WIPO, CIPO, IP Australia, IPOS, the Norwegian, Swedish, and Swiss offices, and INPI France), stratified by filing route (direct versus Madrid designation) and registration era at 500 records per stratum, to exercise legacy-regime rules deliberately.
- **Expanded same-seed sample (stability check).** The same design at tenfold depth: 306,896 records, drawn with the same seed, yielding 282,393 comparable records. Because the seed is identical, this is a strict same-seed superset of the

development sample (every original record is retained), not an independent holdout: the rules were corrected against these same records. It checks whether the headline rates are numerically stable at greater depth, not whether the corrected system generalizes to unseen records.
- **Opposition sample.** 34,132 opposition proceedings with observed filing dates across the five offices whose proceedings data carries oppositions (USPTO, EUIPO, CIPO, IP Australia, IPOS), hash-ordered per office and route cell.

Records with statuses under which no deadline is defined were excluded by pre-registered rule. Because equal-quota strata overweight disagreement-heavy populations (legacy eras, Madrid designations at specific offices) relative to production, the headline is also reported reweighted by each stratum's true production population (9,166,622 records). The weighted Arm A figures and the pooled Arm B and Arm C figures do not yet carry design-based uncertainty intervals; a simple binomial interval would understate the uncertainty because events and comparisons cluster within records and strata, so those intervals are queued as future work.

4.3 Three arms of ground truth

- **Arm A, office-published dates.** For offices that state an expiry or renewal-due date on the record, the computed next renewal deadline must equal the office's stated date exactly. Disagreements are additionally banded by distance. Weekend and holiday filing rollover is treated as a filing-day convention, not an expiry disagreement. The office's stated expiry is withheld from the engine input in this arm, so the computation being tested cannot see the answer (section 3.3).
- **Arm B, event consistency.** For offices whose register histories carry dated renewal and maintenance filings, every observed filing must fall inside the window we computed for it (window open through grace expiry). USPTO events are restricted to a pre-registered allowlist of filing-coded prosecution events, excluding courtesy reminders and office notices, which measure office latency rather than rule correctness.
- **Arm C, observed oppositions.** Every observed opposition must have been filed inside the computed opposition window for the opposed record's publication date, counting the rule's modeled routine extension separately. One non-circular validity filter applies: proceedings data attaches an opposition to both parties' records, and a record already registered when the opposition was filed cannot be the opposed application, so those rows are excluded by registration timing alone.

Two circularity guards matter for reading Arm A. USPTO publishes no expiry date (Signa derives one), so only actually-renewed USPTO records enter Arm A and the USPTO cell is the weakest in the table by construction; the USPTO evidence that matters is Arm B. And where an office publishes a renewal-due date it equals the expiry date in every feed except WIPO, so it is not an independent signal and Arm A uses the stated expiry only.

5. Results

All numbers are reproducible from the committed frozen data and results artifacts; runs are identified by date in the changelog of the evaluation record.

5.1 Headline

Metric	Value	95% CI	n	Dataset
Arm A exact agreement, full population (primary)	96.08%	[96.00, 96.15]	282,393 comparisons	expanded same-seed sample (stability check)
Arm A exact agreement, full population, equal-quota	96.00%	[95.76, 96.22]	29,096	committed small sample
Arm A, full population, production-weighted (pure statutory computation)	98.55%		estimated over a 9.17M-record production population from n=29,096 sampled comparisons	committed small sample
Arm A, modeled population defined after mechanism analysis (post hoc)	99.74%	[99.72, 99.76]	282,393 comparisons	expanded same-seed sample (stability check)
Arm A, modeled population (post hoc)	99.69%	[99.61, 99.75]	27,687	committed small sample
Arm A, production-served schedule, production-weighted (office-date-anchored and therefore partially self-referential; not evidence of rule correctness)	99.89%		estimated over a 9.17M-record production population from n=29,096 sampled comparisons	committed small sample
Arm B event consistency	84.13% pure / 93.14% as served		27,562 events	committed small sample
Arm C opposition consistency	90.23%		23,751 verifiable oppositions	opposition sample
Disagreements classified by mechanism	100%			
Genuinely unexplained residual	0.001%		3 of 282,393	expanded same-seed sample (stability check)

The primary result is full-population agreement: on the population the rules actually compute, without any post-evaluation exclusions, exact agreement is 96.08% at expanded scale (96.00% on the committed small sample). Production-weighting that pure statutory computation by each stratum's true production share gives 98.55%. The 99.74% figure is a secondary result over a modeled population defined after mechanism analysis (post hoc): three subpopulations are excluded whole (agreeing and disagreeing records alike) because a discriminating input is absent from the record (Madrid subsequent designations) or the cadence is a legacy regime with no live forward deadlines (Australian direct marks filed before 1996, Singapore direct marks filed before 1999). These exclusions were defined by inspecting disagreement mechanisms rather than by whether an individual record agreed, but they remain a post hoc estimand change and are labeled as such. The 99.89% "as served" figure adds the office-date anchoring of section 3.3, which is what the API actually returns; because that anchoring uses the office's stated date as an input, it is partially self-referential and is not evidence of rule correctness. The pure-computation figures are the scientifically interesting ones; the as-served figures describe what a docketing customer experiences.

5.2 Arm A by office and route (pure computation, after the section 5.4 fixes)

Pre-registered sample; exact agreement with the office-stated expiry.

Office	Route	n	Exact agreement
EUIPO	direct	2,000	100.00%
EUIPO	madrid	2,000	99.90%
WIPO	madrid (IR)	283	100.00%
Switzerland (IPI)	direct	2,000	99.90%
Switzerland (IPI)	madrid	2,000	99.80%
Sweden (PRV)	direct	1,505	99.93%
Sweden (PRV)	madrid	1,500	99.93%
Norway (NIPO)	direct	2,000	99.35%
Norway (NIPO)	madrid	2,000	99.90%
Canada (CIPO)	direct	2,000	99.65%
Canada (CIPO)	madrid	500	90.60%
France (INPI)	direct	2,000	98.40%
USPTO	direct (derived expiry)	1,502	99.80%
Australia (IPAU)	direct	2,000	84.90%
Australia (IPAU)	madrid	1,500	89.20%
Singapore (IPOS)	direct	2,000	82.75%
Singapore (IPOS)	madrid	1,540	84.29%

The low cells are not scattered error; they are three documented subpopulations. Australian direct marks filed before 1996-01-01 ran seven-year initial and fourteen-year renewal terms under the prior Act, and the register does not carry the date needed to distinguish the roughly one-in-six of that cohort that is genuinely on the modern ladder. Singapore marks filed before

1999-01-15 sat on a comparable repealed-Act cadence with no live forward deadlines. Madrid subsequent designations do not carry their anchoring international-registration date on the row. All three are handled in production by the office-date anchoring of section 3.3, which is why the as-served agreement in those cells reaches 97.1 to 100%. Excluding those three subpopulations whole (agreeing and disagreeing records alike) gives the in-scope headline; the remaining in-scope structural disagreement is 0.20% of comparisons at expanded scale, and every one of those was classified (section 5.5).

5.3 Arm B by office and route (events against windows)

Office	Route	Events	Pure	As served
USPTO	direct	4,426	99.39%	99.39%
USPTO	madrid	1,956	99.59%	99.59%
Switzerland (IPI)	direct	2,014	99.40%	99.40%
Sweden (PRV)	madrid	448	97.54%	97.54%
France (INPI)	direct	2,993	90.11%	90.61%
Sweden (PRV)	direct	5,006	87.91%	88.09%
Canada (CIPO)	direct	2,087	86.97%	87.49%
Australia (IPAU)	direct	6,779	57.62%	90.50%
Australia (IPAU)	madrid	1,830	86.50%	98.25%

The USPTO cells are the cleanest test of the most intricate rules in the system (Section 8, 9, and 71 declaration windows including grace): 99.4% of 6,382 independent filing events fall inside the computed windows. The Australian pure-path cell is the pre-1996 cadence surfacing again; against the office-anchored schedule the same events verify at 90.5%.

5.4 Defects the verification found in our own rules, all fixed

Each fix was verified against primary legal sources and required to fit the register data with no regressions in any other cell. Pre-fix and post-fix agreement is reported for each.

Finding	Statutory basis	Effect of the fix
1. The WIPO configuration was keyed to a different code than production records carry, so every international registration silently received zero computed deadlines	Registry alias, no rule change	766 newly covered IR records agree at 100.00%
2. Canadian legacy renewal cadence: the 2019 reform shortens each term only when the period starts on or after the cutoff, so pre-2019 marks renew on successive 15-year terms first	Trademarks Act (Canada) s.46, in force 2019-06-17	CIPO direct Arm A 73.15% to 99.65%; Arm B 64.73% to 86.97%
3. Madrid designations renew on the international registration's ten-year anniversary, not the national trigger	Madrid Protocol Art. 6(1), 7(1)	CIPO madrid 0.00% to 90.60%; Norway madrid 45.75% to 99.90%; EUIPO madrid 70.30% to 99.90%

Finding	Statutory basis	Effect of the fix
4. The German end-of-month rule leaked onto German Madrid designations	End-of-month rule applies to national registrations only	26 records corrected, up to 30 days
5. The Swedish 2019 trigger reform was keyed on the filing date instead of the registration date, mis-anchoring marks filed before but granted after the reform	Swedish reform of 2019-01-01	5 records corrected by five to seven years
6. US Section 71 windows were anchored on the protection-statement date; the statute keys every Section 71 window to the certificate issuance date, the US registration date	15 U.S.C. 1141k(a); 37 CFR 7.36(b); TMEP 1613.04	USPTO madrid Arm B 66.16% to 99.59%

Finding 6 deserves a note on method: an anchor sweep over 1,956 observed Section 71 filings scored the registration date at 99.39% against 66.16% for the previously used field and at most 64% for every other candidate, and the statutory text, pulled independently, names the same date. Data and statute converged before the rule changed.

5.5 Classification of every remaining disagreement

All 544 in-scope structural disagreements at expanded scale were classified mechanically, each against production event and filing history, with every claimed engine defect adversarially re-verified:

Class	Count	Disposition
Classified by Signa as a register-data error, with per-record evidence	311	The evaluation counts these against us anyway
Anchoring date absent from the record (Madrid subsequent designations, Singapore priority-linked international marks)	225	Handled in production by office-date anchoring
Records mis-routed by ingestion (Madrid designations ingested as direct)	5	Ingestion fix shipped; classification, not rule, error
Genuinely unexplained	3	Open, 0.001% of comparisons

Zero claims survived adversarial verification as a true rule-logic defect: every cohort initially flagged as an engine bug reduced to a missing input or a register error. The register-error class includes French renewals recorded as new deposits under the former law, a Norwegian bulk-load placeholder expiry shared by unrelated marks, and ancient Canadian statutory-anchor phase offsets. Registers are large and contain errors; that a verification of this kind surfaces them, with evidence, is the strongest argument for computing deadlines rather than echoing stored ones.

5.6 Opposition windows against 23,751 observed oppositions

Office	Route	Verifiable n	Consistency
Canada (CIPO)	madrid	1,368	99.85%

Office	Route	Verifiable n	Consistency
EUIPO	direct	4,000	98.22%
EUIPO	madrid	4,000	97.95%
Australia (IPAU)	madrid	1,500	94.80%
Canada (CIPO)	direct	4,000	90.83%
Australia (IPAU)	direct	1,839	90.54%
Singapore (IPOS)	madrid	1,446	87.00%
Singapore (IPOS)	direct	3,991	75.90%
USPTO	direct	1,504	75.27%
USPTO	madrid	103	75.73%
Pooled		23,751	90.23%

This arm found two more defects of the silent-mismatch kind, both fixed. The first is a rule defect (the one opposition-rule defect, which brings the rule-defect total to seven: the six maintenance-rule defects of section 5.4 plus this one); the second is an ingestion defect. (7) The IP Australia opposition rule was keyed to an office code production records do not carry, so every Australian record served a null opposition window until this evaluation noticed. (8) The Singapore connector stored the latest gazette republication as the record's first publication date (the office republishes after amendments and registration), which made 749 sampled oppositions appear to have been filed before publication. Every one of them falls inside the two-month window of the earliest gazette entry; the ingestion fix applies to newly synced records, with a backfill of existing records filed. The evaluation also drove three rule additions verified against it: the Australian window's era split (three months before the 2013-04-15 reform, two after, with the routinely recorded one-month structural extension for the two-step filing process), and dedicated Madrid-designation opposition rules at CIPO, IP Australia, and IPOS, of which only Canada's differs from the domestic rule (a single extension hard-capped at four months by Trademarks Regulations s.125). The Canadian Madrid cell then verifies at 99.85%, with 2 of 1,368 oppositions beyond the statutory cap.

The USPTO cells are depressed by a known data property, not the window arithmetic: proceedings data attaches an opposition to both parties' records, the registration-timing filter can only remove the opposer-side rows it can prove, and the residual splits roughly half extension-boundary cases at the 180-day statutory ceiling and half remaining wrong-side linkage (real, decided oppositions attached to a mark whose single publication was years earlier, which no opposition window can contain). The linkage correction is filed as ingestion work; the Singapore direct cell will be re-measured after the publication-date backfill.

6. Analysis

Misses cluster by mechanism, not randomly. Every material disagreement cohort in this evaluation bucketed cleanly by office, route, or era: a legacy cadence, a wrong anchor column, a keying mismatch, a data-linkage property. Nothing material diffused randomly, which is what one expects if the arithmetic is right and the residual causes are structural. The off-by-N distribution says the same thing: day-level discrepancies are 0.38% of all comparisons; structural misses land months or years off.

The office's date and the statute check each other. The pure computation and the office-stated expiry agree on 99.74% of the modeled population, which means each is a working audit of the other. Where they disagree, adjudication attributed the disagreement to the register more often than to the engine (311 records classified as register-data errors, against no further rule defect identified among the investigated residuals). This is the argument for computing deadlines from statute rather than echoing the register: the computation catches register errors, and the register catches rule errors, and this report is the record of both directions working.

Recomputation at read time beats stored deadlines. Seven rule defects were fixed during this evaluation (six in the maintenance rules, one in the opposition rules). Because deadlines are computed at read time from a pure function, every fix applied to every record instantly, with no re-audit of stored calendars. The before-and-after numbers in section 5.4 were produced by re-running the same frozen sample through the corrected engine, which is exactly what production did the moment each fix deployed.

Arm B is asymmetric, and we score it conservatively. An observed filing inside our window proves the window was usable in practice; the absence of an event proves nothing, and an event recorded outside the window is scored against us whatever its cause. Some of the residual Arm B misses are office recording conventions (renewal events dated at the renewed-until anniversary rather than the filing date), which we deliberately did not exclude.

7. Limitations

- **Coverage is 22 jurisdictions, not all.** Japan, Korea, and China are among the offices whose records Signa carries but whose deadline rules are not yet modeled; the API says `supported: false` for them rather than guessing, and this report claims nothing about them.
- **The USPTO Arm A cell is derived, not office-stated.** USPTO publishes no expiry date, so that one cell shares derivation logic with the thing it tests. The USPTO evidence that matters is Arm B: 99.4% of 6,382 independent filing events inside computed windows.
- **Four offices have no event ground truth.** EUIPO, IPOS, Norway, and WIPO emit no renewal events in our pipeline, so their results rest on Arm A only.
- **Opposition verification is partial where the data is.** The USPTO opposition cells are bounded by a proceedings-to-record linkage property described in section 5.6, and the Singapore direct cell awaits the publication-date backfill. The EUIPO cells (n=8,000, no extension regime to absorb error) are the cleanest external check of the opposition engine.
- **Maintenance-deadline holiday rollover is wired for 10 offices.** Opposition windows apply office holiday calendars across all covered offices; maintenance rollover

uses the calendars where wired and is a small number of days at window edges elsewhere.

- **Some calendar data is forward-looking.** Office holiday calendars are maintained through 2030; a small number of future entries are projected pending official publication and marked as such in the data.
- **One office's renewal trigger is approximated.** The Philippine renewal trigger is computed from registration cycles where the underlying renewal event is not itself carried on the record.
- **The list-endpoint deadline filter is an approximation.** Filtering trademark lists by upcoming deadline uses a stored renewal-due column; the authoritative computation is the detail endpoint and the compute endpoints, which is what this report verifies.
- **Ground truth is the register, and registers contain errors.** That cuts both ways: Arm A disagreements are individually adjudicated rather than assumed to be our fault, and the register-error class in section 5.5 is evidence-backed per record, not a convenient bucket.
- **Rule verification is dated, not continuous, but regression is gated.** Every rule carries the date it was last verified against sources (the configurations touched by this evaluation carry 2026-07-11 or later). Statutes change; the report version will be bumped when rules or measurements are refreshed. Between refreshes, the frozen-sample regression gate of section 3.5 prevents silent movement.

8. Reproducibility and versioning

- The sample manifests (seeded queries and SHA-256 record-identifier digests) were frozen before comparison ran. The evaluation harness, metrics, frozen samples, per-run results, and the mechanism-level classification of every disagreement are committed artifacts, and the headline counts are pinned in continuous integration against the frozen sample.
- The rule corpus this report verifies is publicly inspectable, with statutory citations, via `GET /v1/deadline-rules` and `GET /v1/opposition-rules`.
- The artifacts are published at https://github.com/signa-so/research (folder `deadline-verification`): the frozen manifests, the samples, the events, the oppositions, and the per-record comparison results, so the adjudication of every disagreement can be checked, not taken on faith.
- Source data is public register data; no customer data appears in any evaluation set.
- Published numbers are never edited in place; updated measurements ship as a new version with a changelog entry.

Changelog

- **v0.91 (2026-07-12)**: Framing corrections after an external methodological review. Full-population agreement (96.08% expanded scale, 96.00% committed small sample) is now the primary headline; the 99.74% figure is labeled a post hoc modeled-

population result with its exclusions named, and the 99.89% figure is labeled office-date-anchored and partially self-referential. Removed the "holdout" and "confirmatory" framing (the expanded run is a same-seed superset stability check), narrowed the CI regression-gate claim to what it actually pins, disclosed the 15-to-45-year horizon protocol amendment, reconciled the rule-defect count at seven (six maintenance plus one opposition), softened first-party classification language, added a first-party-evaluation disclosure, and wired the published artifact links.
- **v0.9 (2026-07-12)**: All results populated from the completed evaluation: pre-registered 31,547-record sample, 306,896-record expanded same-seed sample, 34,132-opposition sample. Six maintenance-rule fixes, one opposition-rule fix, two ingestion fixes, and three rule additions documented with pre-fix and post-fix numbers. Engine sections re-verified against the shipped codebase (23 maintenance configurations, 27 opposition rules, 24 holiday calendars, approximately 346 rules-package tests).
- **v0.1-draft (2026-07-07)**: Internal draft. Engine sections verified against the codebase; results pending the frozen evaluation run.
